P4 — Example Optimization Model

Optimal Siting and Sizing of Solar, Wind, and Battery Storage to Compensate for Nuclear Decommissioning in France

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1. Model Structure

The model is formulated as a linear capacity expansion problem over France's 12 administrative regions. The optimization minimizes the annualized cost of renewable generation, battery storage, and transmission expansion while satisfying hourly electricity demand constraints.

The simplified model already includes the main structural elements expected in the final project: multi-region electricity network, hourly energy balance constraints, solar and wind generation using hourly capacity factors, battery storage with state-of-charge dynamics, inter-regional electricity transmission currently modeled with effectively unlimited capacity, and nuclear retirement represented as reduced baseline generation.

The temporal dimension is reduced using 12 representative days (one per month), selected using a k-medoids clustering approach implemented with a 50% weight on electricity demand and the remaining weight distributed across renewable production profiles (see Figure **??**).

Sets and Indices

$$\begin{aligned} r \in \mathcal{R} &= \{1, \dots, 12\} \quad \text{regions} \\ d \in \mathcal{D} &= \{1, \dots, 12\} \quad \text{representative days} \\ t \in \mathcal{T} &= \{1, \dots, 24\} \quad \text{hours per day} \end{aligned}$$

Parameters

$$\begin{aligned} D_{r,d,t} &: \text{electricity demand in region } r \text{ at day } d\text{, hour } t \quad [\text{MW}] \\ \tilde{G}_{r,d,t} &: \text{reduced generation(actual 2024 generation} - \text{retired nuclear)} \quad [\text{MW}] \\ CF^{solar}_{r,d,t} &: \text{solar capacity factor in region } r \text{ at } (d,t) \quad \in [0,1] \\ CF^{wind}_{r,d,t} &: \text{wind capacity factor in region } r \text{ at } (d,t) \quad \in [0,1] \\ c^{solar} &: \text{annualized solar cost} \quad [\$/\text{kW-yr}] \\ c^{wind} &: \text{annualized wind cost} \quad [\$/\text{kW-yr}] \\ c^{bat} &: \text{annualized battery cost (4h)} \quad [\$/\text{kWh-yr}] \\ c^{trans}_{r,r'} &: \text{annualized transmission cost between } r \text{ and } r' \quad [\$/\text{MW-yr}] \\ \overline{T}_{r,r'} &: \text{existing transmission capacity limit between } r \text{ and } r' \quad [\text{MW}] \end{aligned}$$

Decision Variables

$$
\begin{aligned}
x_r^{solar} &\geq 0 : \text{new solar capacity added in region } r \quad \text{[MW]} \\
x_r^{wind} &\geq 0 : \text{new wind capacity added in region } r \quad \text{[MW]} \\
x_r^{bat} &\geq 0 : \text{new battery energy capacity added in region } r \quad \text{[MWh]} \\
x_{r,r'}^{trans} &\geq 0 : \text{new transmission capacity added between } r \text{ and } r' \quad \text{[MW]} \\
f_{r,r',d,t} &: \text{power flow from } r \text{ to } r' \text{ at } (d,t) \quad \text{[MW]} \\
b_{r,d,t}^{+} &\geq 0 : \text{battery charging in region } r \text{ at } (d,t) \quad \text{[MW]} \\
b_{r,d,t}^{-} &\geq 0 : \text{battery discharging in region } r \text{ at } (d,t) \quad \text{[MW]} \\
e_{r,d,t} &\geq 0 : \text{battery state of charge in region } r \text{ at } (d,t) \quad \text{[MWh]}
\end{aligned}
$$

Objective Function

$$
\min \quad \sum_{r\in\mathcal{R}} \left[c^{solar} \cdot x_r^{solar} + c^{wind} \cdot x_r^{wind} + c^{bat} \cdot x_r^{bat} \right] + \sum_{\substack{r,r'\in\mathcal{R} \\ r<r'}} c_{r,r'}^{trans} \cdot x_{r,r'}^{trans} \tag{1}
$$

Constraints

- **Energy Balance (for all r, d, t)**

$$
\tilde{G}_{r,d,t} + CF_{r,d,t}^{solar} \cdot x_r^{solar} + CF_{r,d,t}^{wind} \cdot x_r^{wind} + b_{r,d,t}^{-} - b_{r,d,t}^{+} + \sum_{r'\neq r} f_{r',r,d,t} - \sum_{r'\neq r} f_{r,r',d,t} \geq D_{r,d,t} \tag{2}
$$

- **Transmission Capacity (for all $r \neq r', d, t$)**

$$
f_{r,r',d,t} \leq \overline{T}_{r,r'} + x_{r,r'}^{trans} \tag{3}
$$

$$
-f_{r,r',d,t} \leq \overline{T}_{r,r'} + x_{r,r'}^{trans} \tag{4}
$$

- **Battery State of Charge (for all r, d, t)**

$$
e_{r,d,t} = e_{r,d,t-1} + b_{r,d,t}^{+} - b_{r,d,t}^{-} \tag{5}
$$

- **Battery Initialization — days are independent (for all r, d)**

$$
e_{r,d,0} = 0.5\, x_r^{bat} \tag{6}
$$

- **Battery Energy Capacity (for all r, d, t)**

$$
e_{r,d,t} \leq x_r^{bat} \tag{7}
$$

- **Battery Power Capacity — 4-hour battery (for all r, d, t)**

$$
b_{r,d,t}^{+} \leq \frac{x_r^{bat}}{4}, \qquad b_{r,d,t}^{-} \leq \frac{x_r^{bat}}{4} \tag{8}
$$

- **Non-negativity**

$$
x_r^{solar},\ x_r^{wind},\ x_r^{bat},\ x_{r,r'}^{trans},\ b_{r,d,t}^{+},\ b_{r,d,t}^{-},\ e_{r,d,t} \geq 0 \tag{9}
$$

The final version of the model will additionally include geographic siting constraints, land-use restrictions, explicit transmission expansion costs based on line distances, and more detailed storage modeling.

2. Expected Results

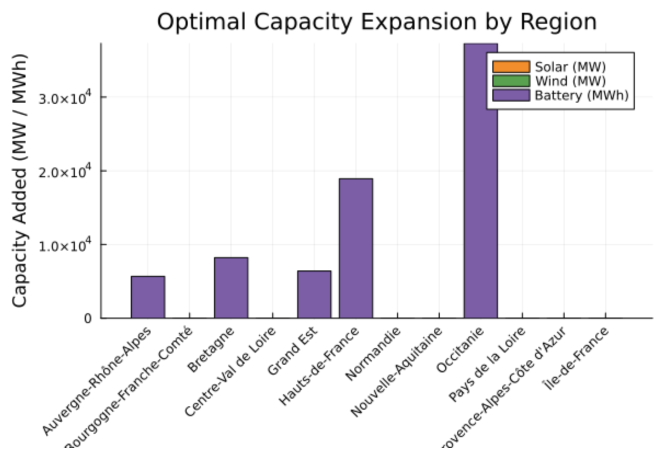
The model is expected to determine the optimal amount and location of solar, wind, and battery capacity additions, evaluate the role of transmission expansion in balancing regional variability, and quantify the tradeoff between storage deployment and grid reinforcement.

In the simplified example case tested so far, transmission was assumed to be effectively unlimited. Under this assumption, the optimization strongly favored battery storage deployment while adding almost no new wind or solar capacity. This behavior is expected because the simplified artificial dataset does not yet include realistic renewable generation variability or transmission congestion. These results are not intended to be physically realistic, but rather to validate the optimization structure and data pipeline.

An example output from the prototype model is shown below:

Region	Solar (MW)	Wind (MW)	Battery (MWh)
Occitanie	0	0	37292
Hauts-de-France	0	0	18923
Bretagne	0	0	8215
Grand Est	0	0	6404

(a) Example optimization results table



(b) Example expected capacity expansion graph

Figure 1: Example outputs from the simplified optimization model with unlimited transmission.

The x-axis represents French administrative regions, while the y-axis shows installed capacity (MW or MWh). Separate bars or curves represent solar, wind, storage, and potentially transmission expansion.

Additional expected figures include hourly supply-demand balance during representative days, battery state-of-charge evolution, regional power flows across the transmission network, and spatial maps of optimal renewable siting.

3. Challenges Encountered

The first challenge was data cleaning and harmonization. Electricity demand, generation, transmission, and weather datasets all come from different sources and use different spatial and temporal resolutions. Significant preprocessing was required to aggregate all data consistently at the regional level.

Another important challenge was reducing the temporal complexity of the optimization problem. Running the model over all 8760 hourly timesteps would be computationally expensive, so representative days were selected using a k-medoids clustering approach. The clustering metric currently gives approximately 50% weight to electricity demand and 50% to renewable generation profiles, weighted by resource type.

Finally, the model was initially infeasible because battery state-of-charge was forced to start at zero at the beginning of each representative day. This prevented the battery from providing flexibility early in the simulation horizon. The issue was resolved by initializing batteries at 50% state of charge instead.